

From Signal Decomposition to Dynamical Decoupling: A Mori–Zwanzig Perspective for UAV Anomaly Detection

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Abstract. Reliable in-flight anomaly detection is critical for UAVs deployed in web-enabled service pipelines, but remains challenging because raw telemetry couples control-induced flight dynamics with disturbance-related residual patterns. Existing UAV-specific methods mainly model temporal deviations and may miss global disturbances, while general time–frequency methods can introduce pseudo-structural spectral noise by directly decomposing control-driven telemetry. To address these issues, we propose *DyDe-AD*, a decomposition-before-detection framework for UAV anomaly detection. *DyDe-AD* first uses a *Mori–Zwanzig-guided Deep Koopman decomposition* to separate each telemetry window into a *Markov-dominant* component $\mathbf{M}$ and a memory-kernel component $\mathbf{P}$, reducing the spectral interference caused by normal control responses. It then reconstructs $\mathbf{M}$ with a *Multi-scale Retrieval-Augmented Temporal Reconstructor (MRA-Rec)* based on scale-specific prototype retrieval, and reconstructs $\mathbf{P}$ with a *Morlet-wavelet variational autoencoder (Morlet-VAE)* for spectral anomaly detection. Experiments on real-world and simulated UAV datasets show that **DyDe-AD** outperforms 17 representative baselines in F1 score and AUC, while using only 0.47 M trainable parameters and achieving the highest inference throughput. The source code is available at <https://anonymous>

Keywords: UAV anomaly detection · Flight telemetry · Dynamic decomposition · Time–frequency modeling · Reconstruction-based detection

1 Introduction

UAV anomaly detection aims to identify abnormal patterns in flight telemetry, such as unexpected attitude variations, irregular control responses, and vibration anomalies, which may indicate system faults, environmental disturbances, or security threats. As UAVs increasingly act as mobile sensing and actuation nodes in web-enabled cyber-physical systems [24], their telemetry streams are continuously collected and analyzed to support real-time decision making. Anomalies in these streams may compromise mission safety and service reliability, making accurate and efficient anomaly detection a critical requirement. With the increasing availability of high-dimensional flight telemetry, data-driven anomaly

detection has become an important research direction for modeling complex UAV behaviors without relying on fully specified physical models.

Existing data-driven UAV anomaly detection methods [2, ?, ?] can model non-linear flight dynamics more flexibly than conventional model-based approaches that rely on accurate system models and predefined fault assumptions [5, ?, ?]. However, most of them focus on temporal dependencies and treat anomalies as local deviations in observation sequences, which limits their ability to detect global disturbances such as wind gusts, rotor degradation, and sensor interference that may induce persistent or periodic variations. Recent general time-series anomaly detection methods mitigate this limitation by complementing temporal modeling with frequency-domain representations, enabling the capture of both transient deviations and global periodic patterns [19, ?, ?, ?]. However, directly applying this time–frequency paradigm to UAV telemetry remains challenging. UAVs are controlled dynamical systems in which flight commands continuously trigger rapid but normal responses in attitude, velocity, and motor states. After frequency-domain decomposition, these legitimate control-induced responses may exhibit broadband spectral patterns similar to anomalous disturbances. As a result, UAV-specific temporal detectors may miss global disturbance evidence, while general time–frequency detectors may confuse normal maneuvers with anomalies. These observations highlight two key challenges.

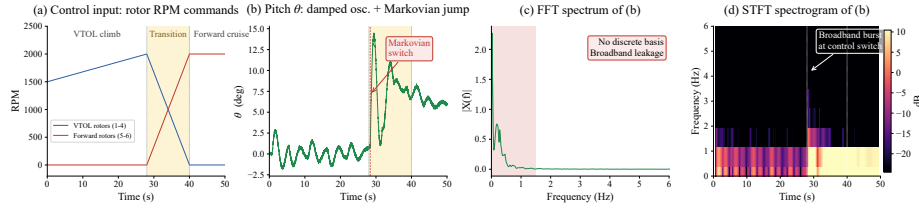


Fig. 1. Pseudo-structural spectral noise caused by direct frequency analysis of UAV flight telemetry.

Challenge 1: Mitigating pseudo-structural spectral noise in frequency analysis. As shown in Figs. 1(a) and (b), control inputs regulate motor RPM and induce abrupt changes in pitch angle θ , reflecting the Markovian nature of UAV control responses. When existing time–frequency methods directly decompose raw telemetry, these instantaneous control-driven transitions are dispersed across a wide frequency range and generate spurious spectral components. This phenomenon is illustrated in Figs. 1(c) and (d), where the FFT spectrum exhibits smeared energy and the STFT spectrogram shows broadband bursts around control-switching instants. Such maneuver-induced artifacts may be confused with fault- or disturbance-related patterns. Therefore, a central problem is how to separate Markovian control dynamics from raw telemetry before applying spectral analysis.

Challenge 2: Reconstructing multi-scale flight dynamics without cross-scale entanglement. UAV flight behaviors naturally evolve at multiple temporal granularities. Fast attitude responses may occur within a few time steps, whereas mode-level transitions, such as hovering, cruising, and loitering, can last for tens or hundreds of steps. Existing reconstruction-based detectors usually encode these heterogeneous patterns into a single shared latent space under flat reconstruction priors. As a result, dynamics from incompatible temporal scales may become entangled, and anomalies occurring at a specific granularity can be partially absorbed by representations optimized for other scales. The key question is how to reconstruct UAV flight behaviors with scale-aware anchors that preserve the intrinsic temporal granularity of different dynamics.

To address these challenges, we propose *DyDe-AD*, a dynamic decomposition and dual-domain anomaly detection framework for UAV telemetry. To reduce *pseudo-structural spectral noise*, we develop a *Mori-Zwanzig-guided Deep Koopman pre-decomposition* strategy that separates raw telemetry into a *Markov-dominant* component $\mathbf{M}$ and an orthogonal *memory-kernel* component $\mathbf{P}$. This decomposition confines abrupt control responses mainly to $\mathbf{M}$ while retaining residual disturbance-related information in $\mathbf{P}$, making subsequent spectral analysis more focused on anomaly-relevant dynamics. To alleviate multi-scale entanglement, we design *MRA-Rec*, a *Multi-scale Retrieval-Augmented Temporal Reconstructor* that reconstructs $\mathbf{M}$ using scale-specific prototypes and scale-indexed cross retrieval. This mechanism constrains reconstruction within appropriate temporal granularities and reduces cross-scale interference. For the memory-kernel component $\mathbf{P}$, we further introduce *Morlet-VAE*, a *Morlet-wavelet-based variational autoencoder* for spectral reconstruction, enabling frequency-domain detection of disturbance-related anomalies.

The main contributions are summarized as follows.

- We propose *DyDe-AD*, a decomposition-before-detection framework for UAV telemetry that separates control-dominated dynamics from disturbance-related residuals and performs anomaly detection in complementary temporal and spectral domains.
- To eliminate pseudo-structural spectral noise, we develop a *Mori-Zwanzig-guided Deep Koopman decomposition* strategy that separates *Markovian* control responses from memory-dependent disturbance patterns before frequency-domain modeling, thereby reducing the spectral interference of normal control action.
- To address multi-scale dynamics entanglement, we design *MRA-Rec* with scale-specific prototype retrieval for temporally granular reconstruction, and introduce *Morlet-VAE* for spectral anomaly detection of the memory-kernel component.
- Extensive experiments against 17 representative baselines demonstrate that *DyDe-AD* achieves superior F1 score and AUC with only 0.47M parameters and high inference throughput, showing its effectiveness and efficiency for real-time UAV anomaly detection.

2 Related Work

UAV Anomaly Detection: From Model-Based Methods to Data-Driven Temporal Modeling. UAV anomaly detection has been studied through both *model-based* and *data-driven* methods. Model-based methods detect anomalies by comparing observed states with predictions from predefined dynamics or sensor models. Extended *Kalman Filter-based* methods [5] use state-estimation residuals, while physics-guided methods, such as *PIPD* [4] and *MARS* [13], incorporate physical constraints, controller information, or aerodynamic signatures. These methods are interpretable, but depend on accurate prior dynamics and fault assumptions, which may be incomplete under wind disturbances, electromagnetic interference, or payload variations.

Data-driven methods reduce this dependency by learning normal telemetry patterns from data. *MeNU* [2] detects anomalies using reconstruction errors, Park et al. [16] exploit *LSTM-based* temporal continuity, and *CLKAN-RF* [9] improves normal-pattern modeling via multi-modal fusion. Although effective for nonlinear temporal dependencies, these methods mainly model observation sequences in the time domain. They are therefore less suited to global disturbances with persistent or periodic patterns, and may absorb command-induced maneuvers into the learned normal space, weakening the distinction between control-dominated dynamics and disturbance-related anomalies.

Time–Frequency Modeling for UAV Anomaly Detection. General time-series anomaly detection methods, such as *Anomaly Transformer* [23], *TranAD* [17], *DCdetector* [25], and *TimesNet* [20], mainly focus on temporal dependency modeling. Recent methods introduce frequency-domain representations to capture global periodic information. *DualTF* [14] extracts spectral features with *FFT*, and *CATCH* [22] models channel correlations along the frequency axis. However, these methods typically apply spectral analysis directly to raw sequences.

Such direct time–frequency modeling is not well aligned with UAV telemetry. Flight commands continuously induce rapid but legitimate responses in attitude, velocity, and motor states. After frequency decomposition, these *Markovian* control responses may spread into broadband spectral artifacts, producing pseudo-structural noise as shown in Figs. 1(c) and (d). Although *TFAD* [27] decomposes time series into trend and periodic components before time–frequency modeling, its decomposition targets generic non-stationarity rather than controlled flight dynamics, and does not explicitly separate control-dominated *Markovian* responses from memory-dependent disturbance patterns.

Overall, UAV-specific methods lack explicit frequency-domain modeling for global disturbance detection, while general time–frequency methods may introduce control-induced spectral artifacts on UAV telemetry. This motivates our decomposition-before-detection framework, which separates control-dominated dynamics from disturbance-related residuals before temporal and spectral anomaly detection.

3 Methodology

3.1 Problem Definition

Definition 1. (*UAV Telemetry Window.*) A UAV telemetry window is represented by $(\mathbf{X}, \mathbf{U})$, where $\mathbf{X} = \{x_t\}_{t=1}^T \in \mathbb{R}^{T \times d_x}$ denotes the state sequence and $\mathbf{U} = \{u_t\}_{t=1}^T \in \mathbb{R}^{T \times d_u}$ denotes the corresponding control sequence within a time horizon of length T .

Problem Definition. (*Decomposition-Based UAV Anomaly Detection.*) Given a training set of normal UAV telemetry windows, the goal is to learn a decomposition function $F : (\mathbf{X}, \mathbf{U}) \mapsto (\mathbf{M}, \mathbf{P})$, where $\mathbf{M}$ denotes the *Markov-dominant* component and $\mathbf{P}$ denotes the memory-kernel component. The anomaly detector models $\mathbf{M}$ and $\mathbf{P}$ with dedicated reconstruction branches to capture control-related flight dynamics and disturbance-related residual patterns, respectively. For a test window, the anomaly decision at time t is given by

$$D(t) = \mathbb{I}[e_M(t) > \tau_M] \vee \mathbb{I}[e_P(t) > \tau_P], \quad (1)$$

where $e_M(t)$ and $e_P(t)$ are the reconstruction errors of the two components, and τ_M and τ_P are their corresponding thresholds. An alarm is raised when either branch detects a significant deviation from the learned normal patterns.

3.2 Framework Overview

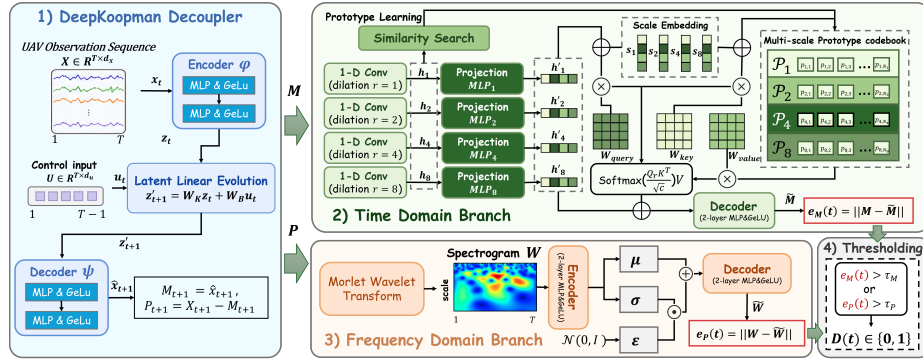


Fig. 2. Overview of the DyDe-AD framework.

Following the *decouple-first, reconstruct-second* design, DyDe-AD consists of three stages: i) DeepKoopman-based sequence decoupling, which leverages a Koopman-driven autoencoder to decouple the UAV observed sequence into a memoryless Markov dominant term $\mathbf{M}_t$ and a memory-kernel term $\mathbf{P}_t$. ii) Multi-scale Retrieval-Augmented Temporal Reconstructor, which first learns scale-indexed prototypes to anchor normal flight regimes at different temporal

granularities, then retrieves the most relevant anchors to reconstruct the Markov dominant term. iii) Wavelet-based Spectral reconstruction, which transforms the memory-kernel term into Morlet wavelet coefficients, reconstructs them with a lightweight VAE. Finally, the discrepancies from both reconstruction branches are aggregated to derive the anomaly score.

3.3 DeepKoopman-based Sequence Decoupling

Preliminary Knowledge. The Mori–Zwanzig (MZ) formalism decomposes the evolution of resolved variables x into three terms:

$$\dot{x}(t) = \underbrace{\Omega x(t)}_{\text{Markov}} + \underbrace{\int_0^t K(t-s) x(s) ds}_{\text{memory kernel}} + \underbrace{f(t)}_{\text{noise}}, \quad (2)$$

where the Markov dominant term captures instantaneous memoryless dynamics, the memory-kernel integral encodes historical accumulation, and $f(t)$ subsumes exogenous perturbations orthogonal to the resolved subspace. The Koopman operator $\mathcal{K}$, defined by $(\mathcal{K}g)(x) = g(\mathcal{T}(x))$, lifts nonlinear state evolution into a linear operator on the space of observables. Lin et al. [10] prove that a neural-network instantiation of $\mathcal{K}$ (DeepKoopman) can theoretically approximate the Markov dominant term of the MZ decomposition: the linear latent transition $W_K z_t$ corresponds to $\Omega x(t)$, while the reconstruction residual naturally absorbs the memory-kernel and noise terms.

Rather than directly applying time–frequency analysis to the raw UAV sequence, this stage decomposes the raw sequence into two components for branch-specific reconstruction: a Markov dominant term for the temporal branch and a memory-kernel term for the spectral branch. This avoids projecting normal control-induced responses into frequency components, which would otherwise cause legitimate maneuvers to be misidentified as anomalies. To achieve this, we instantiate the Mori–Zwanzig decomposition with a Koopman-driven autoencoder. As shown in Fig. 2, it maps the sequence into a latent space, predicts the memoryless Markov dynamics through a control-aware linear transition, and uses the decoded prediction and its residual as the Markov dominant term and the memory-kernel term, respectively.

Specifically, given the observed UAV sequence $\mathbf{X} \in \mathbb{R}^{T \times d_x}$, we use a 2-layer MLP encoder φ to map each state $x_t \in \mathbb{R}^{d_x}$ into a latent representation $z_t = \varphi(x_t) \in \mathbb{R}^c$. In the latent space, the *Markov* component is assumed to depend only on the current latent state and the current control input. Thus, we model the one-step control-aware transition as

$$z'_{t+1} = W_K z_t + W_B u_t, \text{ where } z'_1 = z_1 \quad (3)$$

where $W_K \in \mathbb{R}^{c \times c}$ and $W_B \in \mathbb{R}^{c \times d_u}$ are learnable matrices. The predicted latent state is decoded by another 2-layer MLP ψ to obtain the Markov dominant prediction $\hat{x}_{t+1} = \psi(z'_{t+1})$. The predictions over the whole window form the

Markov dominant term $\mathbf{M} \in \mathbb{R}^{T \times c}$ for temporal reconstruction, while the residual $p_{t+1} = x_{t+1} - \hat{x}_{t+1}$ forms the memory-kernel term $\mathbf{P} \in \mathbb{R}^{T \times c}$ for spectral reconstruction.

3.4 Multi-scale Retrieval-Augmented Temporal Reconstructor

After sequence decoupling, the *Markov* dominant term $\mathbf{M}$ is fed into the temporal reconstruction branch. This branch models normal UAV flight dynamics and reconstructs $\mathbf{M}$ accordingly. During inference, deviations from the learned normal dynamics are expected to produce larger reconstruction errors, which serve as temporal anomaly evidence.

To satisfy the lightweight and low-latency requirements of UAV anomaly detection, we adopt an *autoencoder-based reconstructor* instead of a *Transformer-based sequence model*. However, $\mathbf{M}$ contains flight patterns at multiple temporal granularities, ranging from rapid control responses to longer maneuver-level regimes. A naive autoencoder with a single continuous latent space may oversmooth these heterogeneous normal patterns and weaken anomaly discrimination. Therefore, we extend the vector-quantised codebook [18] to scale-indexed prototypes, which provide discrete anchors for normal flight regimes, while retrieval supplies scale-specific normal context for reconstruction.

Multi-scale Sequential Encoding. We leverage 1-D convolutional encoders with different receptive fields to extract temporal features from the Markov dominant term $\mathbf{M}$ along the sequential dimension. Specifically, we build four parallel convolutional branches with kernel size 3 and dilation rates $r \in \mathcal{R}$, $\mathcal{R} = \{1, 2, 4, 8\}$, where each branch encodes $\mathbf{M}$ into a scale-specific representation:

$$\mathbf{H}_r = \text{Conv}_r^{k=3}(\mathbf{M}), \quad \mathbf{H}_r \in \mathbb{R}^{T \times c}, \quad r \in \{1, 2, 4, 8\}, \quad (4)$$

The small-dilation branch captures short-term control responses, while larger-dilation branches capture maneuver-level variations and longer flight-regime patterns. These multi-scale representations provide the temporal context for the subsequent prototype retrieval and reconstruction.

Multi-scale Prototype Learning. For each temporal scale $r \in \mathcal{R}$, we maintain an independent prototype codebook $\mathcal{P}_r = \{\mathbf{p}_{r,1}, \dots, \mathbf{p}_{r,N_r}\} \in \mathbb{R}^{N_r \times c}$, where each prototype represents a recurring normal flight pattern under the corresponding temporal granularity. Given $\mathbf{H}_r = [\mathbf{h}_{r,1}, \dots, \mathbf{h}_{r,T}] \in \mathbb{R}^{T \times c}$, each temporal feature is assigned to its nearest prototype:

$$q_{r,t} = \arg \min_{j \in \{1, \dots, N_r\}} \|\mathbf{h}_{r,t} - \mathbf{p}_{r,j}\|_2^2, \quad \mathbf{e}_{r,t} = \mathbf{p}_{r,q_{r,t}}. \quad (5)$$

This produces the quantized prototype sequence $\mathbf{E}_r = [\mathbf{e}_{r,1}, \dots, \mathbf{e}_{r,T}] \in \mathbb{R}^{T \times c}$, which replaces continuous temporal features with discrete normal-pattern anchors.

The prototype codebook is learned jointly with the temporal reconstructor. During training on normal flight data, each feature $\mathbf{h}_{r,t}$ is encouraged to stay

close to its assigned prototype, while each selected prototype is updated to represent the normal patterns assigned to it. This is implemented with a quantization loss that pulls the continuous features and their selected prototypes toward each other:

$$\mathcal{L}_{\text{vq}}^r = \sum_{t=1}^T \|\text{sg}[\mathbf{h}_{r,t}] - \mathbf{e}_{r,t}\|_2^2 + \eta \sum_{t=1}^T \|\mathbf{h}_{r,t} - \text{sg}[\mathbf{e}_{r,t}]\|_2^2, \quad (6)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation and η controls the commitment strength. The first term updates the selected prototypes toward the encoder features, while the second term encourages the encoder output to remain close to the selected prototypes. The total prototype learning loss is $\mathcal{L}_{\text{vq}} = \sum_{r \in \mathcal{R}} \mathcal{L}_{\text{vq}}^r$.

By learning a separate codebook for each temporal scale, the model obtains scale-specific discrete anchors for normal UAV flight regimes. Fine-scale codebooks capture short control-induced variations, whereas larger-scale codebooks capture maneuver-level and regime-level patterns.

Scale-indexed Cross Retrieval. After obtaining the scale-specific feature sequence $\mathbf{H}_r$, we first feed it into an MLP projection layer to obtain $\mathbf{H}'_r = \text{MLP}_r(\mathbf{H}_r) \in \mathbb{R}^{T \times c}$. To make the retrieval process aware of temporal granularity, we introduce a learnable scale embedding $\mathbf{s}_r \in \mathbb{R}^c$ for each scale $r \in \mathcal{R}$. The query is constructed from the projected feature and its corresponding scale embedding as $\mathbf{Q}_r = (\mathbf{H}'_r + \mathbf{1}_T \mathbf{s}_r^\top) \mathbf{W}_{\text{query}}$, where $\mathbf{W}_{\text{query}} \in \mathbb{R}^{c \times c}$ and $\mathbf{1}_T$ denotes an all-one vector of length T . For the prototype library, we add the corresponding scale embedding to each prototype codebook before forming keys, so that prototypes from different temporal granularities can be distinguished during retrieval:

$$\mathbf{K} = \text{Concat}_{r' \in \mathcal{R}} [(\mathcal{P}_{r'} + \mathbf{1}_{N_{r'}} \mathbf{s}_{r'}^\top) \mathbf{W}_{\text{key}}], \quad \mathbf{V} = \text{Concat}_{r' \in \mathcal{R}} [\mathcal{P}_{r'} \mathbf{W}_{\text{value}}], \quad (7)$$

where $\mathbf{W}_{\text{key}}, \mathbf{W}_{\text{value}} \in \mathbb{R}^{c \times c}$, and $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times c}$ with $N = \sum_{r' \in \mathcal{R}} N_{r'}$. We then perform scale-aware cross attention between the scale-specific query and the multi-scale prototype library:

$$\mathbf{R}_r = \text{softmax} \left(\frac{\mathbf{Q}_r \mathbf{K}^\top}{\sqrt{c}} \right) \mathbf{V}, \quad \tilde{\mathbf{H}}_r = \mathbf{H}'_r + \mathbf{R}_r. \quad (8)$$

The retrieved prototype representation $\mathbf{R}_r$ provides normal flight-regime context selected from the multi-scale library, and $\tilde{\mathbf{H}}_r$ is the retrieval-enhanced temporal representation at scale r .

Reconstruction and Anomaly Scoring. After scale-aware prototype retrieval, we obtain the enhanced representations $\{\tilde{\mathbf{H}}_r\}_{r \in \mathcal{R}}$ from different temporal granularities. We concatenate them along the feature dimension and feed the fused representation into a lightweight 2-layer MLP-based decoder D_T to reconstruct the Markov dominant term:

$$\tilde{\mathbf{H}} = \text{Concat}_{r \in \mathcal{R}} (\tilde{\mathbf{H}}_r) \in \mathbb{R}^{T \times 4c}, \quad \tilde{\mathbf{M}} = D_T(\tilde{\mathbf{H}}) \in \mathbb{R}^{T \times d}. \quad (9)$$

The temporal reconstruction discrepancy is then computed between the input Markov dominant term and its reconstruction. For each timestamp t , we define

the temporal discrepancy as $e_M(t) = \left\| \mathbf{M}_t - \widetilde{\mathbf{M}}_t \right\|_2^2$, this discrepancy measures whether the Markov dominant dynamics can be explained by the learned normal flight-regime prototypes.

3.5 Wavelet-based Spectral Reconstruction

In parallel with the temporal reconstruction branch, the frequency reconstruction branch reconstructs the memory-kernel term $\mathbf{P}$. This term contains residual dynamics that cannot be explained by the memoryless Markov component, such as exogenous disturbances and accumulated perturbations over time. Since different UAV faults may appear in different frequency bands, $\mathbf{P}$ usually spans a wide frequency range and is more suitable to be represented in the frequency domain.

Specifically, we apply the Morlet wavelet transform to $\mathbf{P}$. For each channel, the wavelet coefficient at scale s and temporal location τ is computed as

$$W(s, \tau) = \int p(t) \frac{1}{\sqrt{s}} \phi^* \left(\frac{t - \tau}{s} \right) dt, \quad (10)$$

where $\phi(\cdot)$ denotes the Morlet mother wavelet. Applying this transform over all channels produces a spectrum map $\mathbf{W} \in \mathbb{R}^{T \times S \times d}$, where S denotes the number of wavelet scales.

The spectrum map is then reconstructed by a lightweight VAE. Both the encoder and decoder of the VAE are implemented as 2-layer MLPs. The encoder maps $\mathbf{W}$ into a latent distribution $q_\theta(\mathbf{z} \mid \mathbf{W})$, and the decoder reconstructs the spectrum map as $\widetilde{\mathbf{W}}$. For each timestamp t , we compute the spectral discrepancy by comparing the original and reconstructed spectrum maps across all wavelet scales:

$$e_P(t) = \left\| \mathbf{W}_{t,:,:} - \widetilde{\mathbf{W}}_{t,:,:} \right\|_2^2. \quad (11)$$

This discrepancy measures whether memory-kernel term conforms to the normal time-frequency distribution learned from normal flight windows.

3.6 Training and Inference

Two-Stage Training. Jointly training the decoupling module and the two reconstruction branches is unstable, because reconstruction losses may change the latent space and weaken the intended Markov/memory-kernel separation. We therefore adopt a two-stage training strategy.

In the first stage, we train the DeepKoopman-based sequence decoupling module, including the encoder φ , decoder ψ , and the linear transition matrices W_K and W_B , with

$$\mathcal{L}_{\text{stage1}} = \alpha \sum_{t=1}^T \|\psi(\varphi(x_t)) - x_t\|_2^2 + \beta \sum_{t=1}^{T-1} \|z_{t+1} - (W_K z_t + W_B u_t)\|_2^2. \quad (12)$$

The first term preserves the reconstruction ability of the autoencoder, while the second term enforces the one-step Markov evolution in the latent space. After this stage, the observed sequence is decomposed into a Markov dominant term $\mathbf{M}$ and a memory-kernel term $\mathbf{P}$.

In the second stage, we freeze the decoupling module and jointly train the temporal and frequency reconstruction branches on the fixed $\mathbf{M}$ and $\mathbf{P}$. The temporal branch optimizes the Markov-term reconstruction loss together with the scale-indexed prototype learning objective, while the retrieval module, including the scale embeddings and attention projections $\mathbf{W}_{query}, \mathbf{W}_{key}, \mathbf{W}_{value}$, is optimized through the temporal reconstruction objective. The frequency branch reconstructs the Morlet spectrum map and optimizes the VAE objective. The Stage-2 objective is

$$\mathcal{L}_{\text{stage2}} = \underbrace{\|\mathbf{M} - \widetilde{\mathbf{M}}\|_2^2 + \lambda_{\text{vq}} \mathcal{L}_{\text{vq}}}_{\mathcal{L}_{\text{temp}}} + \lambda_{\text{freq}} \underbrace{\left(\|\mathbf{W} - \widetilde{\mathbf{W}}\|_2^2 + \text{KL}(q_{\theta}(\mathbf{z} | \mathbf{W}) \| p(\mathbf{z})) \right)}_{\mathcal{L}_{\text{freq}}}, \quad (13)$$

where $\mathcal{L}_{\text{vq}} = \sum_{r \in \{1, 2, 4, 8\}} \mathcal{L}_{\text{vq}}^r$ updates the scale-indexed prototype libraries and encourages temporal features to stay close to their assigned prototypes.

Inference. During inference, a test sequence is first decomposed into $\mathbf{M}$ and $\mathbf{P}$ by the trained *DeepKoopman-based* decoupling module. The temporal branch reconstructs $\mathbf{M}$ to produce $e_M(t)$, and the spectral branch reconstructs the Morlet-wavelet spectrum of $\mathbf{P}$ to produce $e_P(t)$. Following the problem definition, a timestamp is flagged as anomalous if either branch exceeds its validation-set threshold, as defined in Eq. (1).

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate *DyDe-AD* on two UAV anomaly detection datasets. *Rfly-MAD* [8] contains 5,629 real and simulated multirotor flights with 10 anomaly types, including motor faults, propeller faults, and wind effects. *ALFA* [6] contains 47 real fixed-wing flight cases with 8 fault types, such as aileron and engine failures. For both datasets, multi-sensor signals are resampled to 100 Hz. Continuous signals below 100 Hz are linearly interpolated, while those above 100 Hz are downsampled by mean aggregation within each 10 ms interval. Discrete control signals are forward-filled or aggregated by median. Models are trained only on normal flight cases, while anomalous cases are used for validation and testing. The task is to distinguish pre-fault normal windows from post-fault anomalous windows within each anomalous flight.

Baselines. We compare *DyDe-AD* with 17 representative baselines from five categories: traditional methods, including *LOF* [1], *Isolation Forest* [11], *HBOS* [3], and *EKF* [5]; autoencoder-based methods, including *VAE* [7], *ContraAD* [30], and *KAN-AD* [29]; Transformer-based methods, including *TranAD* [17], *PatchTST* [15], *iTransformer* [12], *Autoformer* [21], and *Informer* [28]; linear

models, including *DLinear* [26]; and time-frequency methods, including *TimesNet* [20], *DualTF* [14], *CATCH* [22], and *TFAD* [27]. These baselines cover major paradigms in time-series anomaly detection and UAV anomaly detection.

Evaluation Metrics. We report Precision, Recall, F1 score, and AUC under strict point-wise evaluation without Point-Adjustment. Precision (P) measures the reliability of detected anomalies, Recall (R) measures the coverage of true anomalies, F1 summarizes their balance, and AUC evaluates threshold-independent discrimination. All results are reported as mean $\pm$ standard deviation over five folds.

Implementation Details. We use stratified 5-fold cross-validation over flight cases rather than windows, ensuring that windows from the same trajectory do not appear in different splits. Thresholds are selected on the validation fold by Best-F1 search over a $[0\%, 25\%]$ anomaly-rate range. For *DyDe-AD*, the dual thresholds (τ_M, τ_P) are jointly searched on a 25×25 grid. Unless otherwise specified, the sliding window length is $W=175$ with stride $\sigma=5$, and the *Koopman latent dimension* is set to approximately $1.5d_x$ (64 for *RflyMAD* and 48 for *ALFA*). The encoder φ and decoder ψ are two-layer *MLPs* with *GELU* activations. *MRA-Rec* uses parallel dilated encoders with rates $\{1, 2, 4, 8\}$ and 32 prototypes per scale. The model is optimized with *Adam* using a cosine-annealed learning rate from 1×10^{-3} to 1×10^{-5} , batch size 128, and 100/50 epochs for Stage 1/Stage 2. Early stopping is applied after 3 stagnant validation epochs. Experiments are conducted on an NVIDIA V100S GPU with PyTorch 2.8.0 and CUDA 12.6.

4.2 Overall anomaly detection performance

Table 1. Point-wise anomaly detection on RflyMAD and ALFA against **17 baselines**. **Bold:** best; underlined: second-best. AUC with four decimals; P/R/F1 in %.

Dataset Metric	RflyMAD				ALFA			
	P	R	F1	AUC	P	R	F1	AUC
LOF [1]	87.04 $\pm$ 3.79	60.29 $\pm$ 2.24	71.13 $\pm$ 0.58	0.8687 $\pm$ 0.0036	58.43 $\pm$ 1.13	58.58 $\pm$ 1.23	58.52 $\pm$ 0.94	0.5342 $\pm$ 0.0595
IsolationForest [11]	68.21 $\pm$ 1.48	81.41 $\pm$ 2.37	74.20 $\pm$ 1.05	0.8012 $\pm$ 0.0030	63.65 $\pm$ 4.67	54.16 $\pm$ 7.27	57.96 $\pm$ 2.56	0.5422 $\pm$ 0.0465
HBOS [3]	66.60 $\pm$ 3.69	72.43 $\pm$ 5.13	69.12 $\pm$ 1.14	0.7699 $\pm$ 0.0076	48.62 $\pm$ 5.59	67.66$\pm$7.89	55.87 $\pm$ 3.88	0.4487 $\pm$ 0.0676
VAE [7]	83.35 $\pm$ 1.59	91.58 $\pm$ 1.05	87.27 $\pm$ 0.31	0.9128 $\pm$ 0.0010	61.88 $\pm$ 2.84	56.94 $\pm$ 4.84	59.35 $\pm$ 1.43	0.6284 $\pm$ 0.0413
EKF [5]	87.37 $\pm$ 2.97	87.19 $\pm$ 2.61	87.21 $\pm$ 0.23	0.9517 $\pm$ 0.0017	70.64 $\pm$ 4.55	56.36 $\pm$ 3.24	62.39 $\pm$ 1.18	0.6401 $\pm$ 0.0283
TranAD [17]	82.89 $\pm$ 1.35	93.10 $\pm$ 0.48	87.57 $\pm$ 0.78	0.9373 $\pm$ 0.0031	69.29 $\pm$ 4.23	59.86 $\pm$ 4.31	<u>64.23$\pm$2.73</u>	0.6085 $\pm$ 0.0372
DLinear [26]	65.42 $\pm$ 0.75	79.67 $\pm$ 0.54	71.99 $\pm$ 0.53	0.5581 $\pm$ 0.0037	59.48 $\pm$ 2.90	39.55 $\pm$ 1.66	47.51 $\pm$ 1.03	0.5032 $\pm$ 0.0359
PatchTST [15]	54.90 $\pm$ 0.84	95.31 $\pm$ 0.81	69.67 $\pm$ 0.58	0.5584 $\pm$ 0.0029	60.95 $\pm$ 2.56	37.66 $\pm$ 1.20	46.56 $\pm$ 0.91	0.5150 $\pm$ 0.0385
TimesNet [20]	65.90 $\pm$ 0.48	94.23 $\pm$ 1.15	77.57 $\pm$ 0.54	0.6620 $\pm$ 0.0033	58.66 $\pm$ 4.65	47.25 $\pm$ 3.89	52.34 $\pm$ 1.86	0.4846 $\pm$ 0.0405
Autoformer [21]	81.44 $\pm$ 1.92	88.17 $\pm$ 0.90	84.67 $\pm$ 1.06	0.9296 $\pm$ 0.0038	69.50 $\pm$ 4.53	52.01 $\pm$ 2.90	59.50 $\pm$ 0.81	0.6210 $\pm$ 0.0395
Informer [28]	86.72 $\pm$ 1.55	88.77 $\pm$ 1.31	87.72 $\pm$ 0.31	<u>0.9570$\pm$0.0009</u>	69.31 $\pm$ 5.13	55.38 $\pm$ 4.41	61.57 $\pm$ 1.48	0.6225 $\pm$ 0.0448
iTransformer [12]	56.29 $\pm$ 0.62	73.42 $\pm$ 2.25	63.71 $\pm$ 1.09	0.5019 $\pm$ 0.0033	59.18 $\pm$ 2.79	39.26 $\pm$ 1.48	47.21 $\pm$ 0.97	0.5003 $\pm$ 0.0454
ContraAD [30]	54.24 $\pm$ 0.82	98.02$\pm$0.43	69.83 $\pm$ 0.69	0.6475 $\pm$ 0.0067	59.20 $\pm$ 5.21	54.64 $\pm$ 3.30	56.83 $\pm$ 2.07	0.4899 $\pm$ 0.0557
DualTF [14]	87.79 $\pm$ 0.69	96.61 $\pm$ 0.83	<u>91.99$\pm$0.34</u>	0.9564 $\pm$ 0.0015	<u>70.99$\pm$3.81</u>	52.03 $\pm$ 5.01	60.05 $\pm$ 2.90	0.6476 $\pm$ 0.0257
KAN-AD [29]	76.16 $\pm$ 1.35	<u>97.51$\pm$0.36</u>	85.52 $\pm$ 1.04	0.7875 $\pm$ 0.0026	66.69 $\pm$ 3.67	49.38 $\pm$ 2.09	56.74 $\pm$ 0.78	0.5983 $\pm$ 0.0212
CATCH [22]	86.79 $\pm$ 0.90	95.10 $\pm$ 1.95	90.74 $\pm$ 0.58	0.9488 $\pm$ 0.0018	71.13$\pm$3.95	54.98 $\pm$ 2.83	62.02 $\pm$ 1.49	<u>0.6510$\pm$0.0428</u>
TFAD [27]	86.77 $\pm$ 0.87	94.35 $\pm$ 0.91	90.42 $\pm$ 0.63	0.9301 $\pm$ 0.0027	67.53 $\pm$ 3.13	67.98 $\pm$ 2.72	62.18 $\pm$ 1.34	0.6497 $\pm$ 0.0317
DyDe-AD (Ours)	92.21$\pm$0.96	93.73 $\pm$ 0.74	92.96$\pm$0.23	0.9585$\pm$0.0017	68.99 $\pm$ 3.40	<u>63.84$\pm$3.19</u>	66.32$\pm$1.65	0.6741$\pm$0.0384

Table 1 reports the point-wise mean $\pm$ std results over five folds. *DyDe-AD* achieves the best F1 score and AUC on both datasets. On *RflyMAD*, it improves over the strongest competing results by +0.97 F1 and +0.0021 AUC. On *ALFA*, the corresponding improvements are +2.09 F1 and +0.0231 AUC. These results indicate that *DyDe-AD* is effective across both multirotor and fixed-wing UAV scenarios.

The performance gains are mainly attributed to two design choices. First, the *Mori-Zwanzig-guided Deep Koopman decomposition* separates the *Markov-dominant* component from the memory-kernel component, reducing the interference between control-induced dynamics and disturbance-related residuals. Second, the dual-branch reconstruction design models the two components with different inductive biases, where *MRA-Rec* captures multi-scale temporal flight dynamics and *Morlet-VAE* captures spectral disturbance patterns.

Compared with single-domain baselines such as *CATCH*, *TranAD*, and *Informers*, *DyDe-AD* can jointly characterize abrupt temporal deviations and global spectral variations. Compared with *DualTF*, which directly models raw sequences in both time and frequency domains, *DyDe-AD* performs decomposition before time-frequency modeling. This design reduces pseudo-structural spectral noise caused by normal control responses and leads to more reliable anomaly evidence in the temporal and spectral branches.

4.3 Model Analyze

Ablation Study. To evaluate the contribution of each component, we compare *DyDe-AD* with six variants:

- **w/o Decoupling:** removes the Deep Koopman-based decomposition and feeds raw $\mathbf{X}$ to both branches.
- **w/o MRA-Rec:** removes the Markov-dominant branch and relies only on the memory-kernel branch.
- **w/o Morlet-VAE:** removes the memory-kernel branch and relies only on the Markov-dominant branch.
- **w/o Two-Stage Training:** jointly optimizes decomposition and reconstruction in a single stage.
- **Dual-MRA-Rec:** applies MRA-Rec to both decomposed components.
- **Dual-Morlet-VAE:** applies Morlet-VAE to both decomposed components.

Fig. 3(a) shows that all variants perform worse than the full model. Removing decoupling leads to clear degradation, indicating that directly modeling raw telemetry introduces interference between control-induced dynamics and disturbance-related residuals. Removing *MRA-Rec* or *Morlet-VAE* also reduces performance, showing that temporal reconstruction of the *Markov-dominant* component and spectral reconstruction of the memory-kernel component are both necessary. The performance drops of *w/o Two-Stage Training*, *Dual-MRA-Rec*, and *Dual-Morlet-VAE* further suggest that *DyDe-AD* benefits from preserving the intended decomposition structure and using branch-specific reconstruction models.

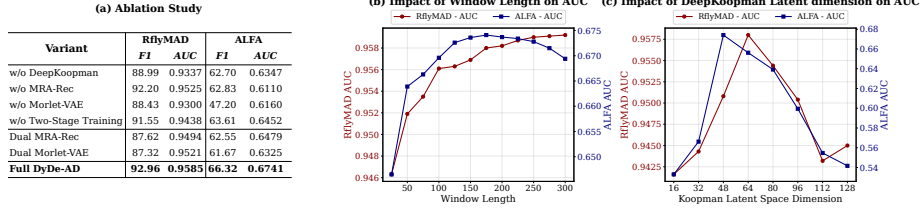


Fig. 3. Ablation study and hyperparameter sensitivity.

Hyperparameter Sensitivity. We study the sensitivity of *DyDe-AD* to the window length W and the Koopman latent dimension d_z .

Window length. We vary W from 25 to 300 with a step size of 25. As shown in Fig. 3(b), performance is limited when the window is too short to cover sufficient flight context. It improves as the window captures more complete control-response patterns and peaks around $W=175$ on both datasets. Longer windows may introduce unrelated dynamics and weaken decomposition quality.

Latent dimension. Fig. 3(c) shows that performance is best when d_z is approximately $1.5d_x$. Smaller latent spaces may lose useful flight information, while larger ones may introduce redundant dimensions and weaken the separation between *Markovian* and memory-dependent dynamics.

Efficiency Analysis. We evaluate efficiency using model size, peak GPU memory, and inference throughput. As shown in Fig. 4, *DyDe-AD* uses 0.47 M parameters, requires about 210 MB peak GPU memory, and achieves the highest inference throughput among all baselines. This efficiency is mainly due to the compact decomposition design, which reduces reconstruction complexity by separating telemetry into two lower-complexity components. The accuracy–latency Pareto results in Figs. 5(a) and (b) further show that *DyDe-AD* achieves a favorable trade-off between detection performance and computational cost.

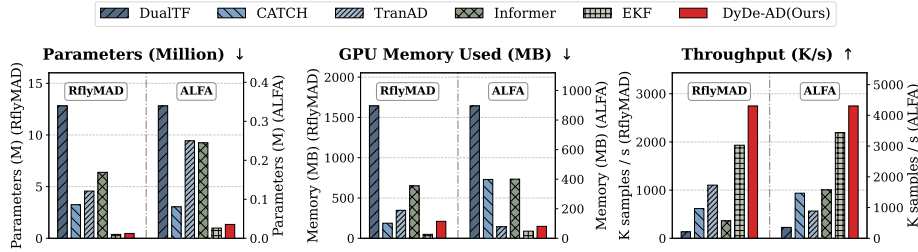


Fig. 4. Efficiency comparison with baselines.

Error-Space Analysis. Fig. 5(c) shows the kernel-density estimates of (e_M, e_P) for three *RflyMAD* fault groups, including dynamic component faults,

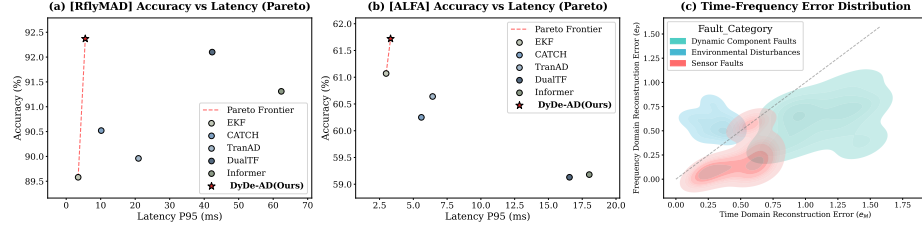


Fig. 5. Accuracy–efficiency trade-off and error-space distribution.

sensor faults, and environmental disturbances. These groups exhibit distinct distributions in the two-branch error space. Dynamic component faults tend to induce larger e_M , environmental disturbances lead to higher e_P , and sensor faults are mostly located near the low-error region. This pattern is consistent with the intended decomposition of *DyDe-AD*, where the *Markov-dominant branch* captures abnormal flight dynamics and the memory-kernel branch captures disturbance-related residual patterns.

5 Conclusion

We investigated the limitations of existing UAV telemetry anomaly detection methods and identified two key challenges, namely pseudo-structural spectral noise caused by control-induced dynamics and multi-scale entanglement in reconstruction modeling. To address these challenges, we proposed *DyDe-AD*, a *decomposition-before-detection* framework that separates telemetry into a *Markov-dominant* component and a memory-kernel component, followed by *branch-specific reconstruction* in the temporal and spectral domains. The *Mori-Zwanzig-guided Deep Koopman decomposition* reduces control-induced spectral interference, while *MRA-Rec* and *Morlet-VAE* model multi-scale flight dynamics and disturbance-related spectral patterns, respectively. Experiments on *RflyMAD* and *ALFA* show that *DyDe-AD* outperforms 17 representative baselines in F1 score and AUC with only 0.47M trainable parameters and high inference throughput. Additional ablation, sensitivity, efficiency, and error-space analyses further validate the effectiveness of dynamic decomposition and branch-specific reconstruction. Future work will extend *DyDe-AD* to heterogeneous UAV fleets and online adaptation under non-stationary flight environments.

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